

1.0 THE SATELLITE SYSTEM CONFIGURATION

This is a vast subject, and one that has provided the material for many texts since its inception in 1962. An exhaustive coverage of satellite communications is impossible within a few pages, and in this text an attempt will be made to present only the main principles involved. Some of the basic engineering design aspects will be discussed, and how the laws of physics lead to specific equipment configurations.

There are three categories of communications satellite:

1. Fixed service
2. Broadcast service
3. Mobile service

The fixed service type is associated with geostationary satellite orbit (GSO) systems for international and sometimes domestic traffic. Very-small-aperture terminal (VSAT) systems are fixed services that use 1-m or so diameter antennas, and provide the important platform for two-way data communications.

Broadcast services provide distribution of TV to inexpensive receive-only terminals using antennas as small as about a half meter (18 in) in diameter.

Mobile services initially used GSO systems for applications such as maritime or news gathering. The new LEO and MEO systems provide telephony and limited data to individual pocket phones.

In many respects the satellite communication link can be viewed as a super-long-distance microwave link, and many of the calculations for terrestrial links can be extended to satellite paths. As with terrestrial telecommunication systems, the satellite industry is gradually becoming digitalized. Interestingly, the satellite is transparent to the passage of analog or digital information. Also, the baseband signals, whether analog or digital, are packaged on an analog radio carrier and only very recent satellites incorporate demodulation or demultiplexing. Some satellites already have onboard processing that is done at the individual channel (subbaseband) level, but the increased level of complexity and cost-effectiveness of future onboard processing is still being debated.

Regardless of whether satellite communications are international or domestic, there are several major categories of satellite users:

1. Individual voice band
2. Corporate data
3. TV broadcasting
4. Government and military

Satellite systems are very attractive for satisfying the wide area of coverage and the point-to-multipoint nature required for broadcasting. Video coverage of an event at one place on the globe can be sent up to a satellite and redistributed (broadcast) over large areas of the populated world in the form of clear television pictures. A network of satellites can provide global coverage so that telephone conversations can take place between individuals located at any of the remotest places in the world, with only a pocket handset required by the participants.

Government applications are mainly to provide surveillance information, and military operations are coordinated and facilitated by satellite communications.

Each application requires a significantly different system design approach. Perhaps the most serious limitation of satellite communications is the *total available satellite bandwidth*. Although the information-carrying capacity of

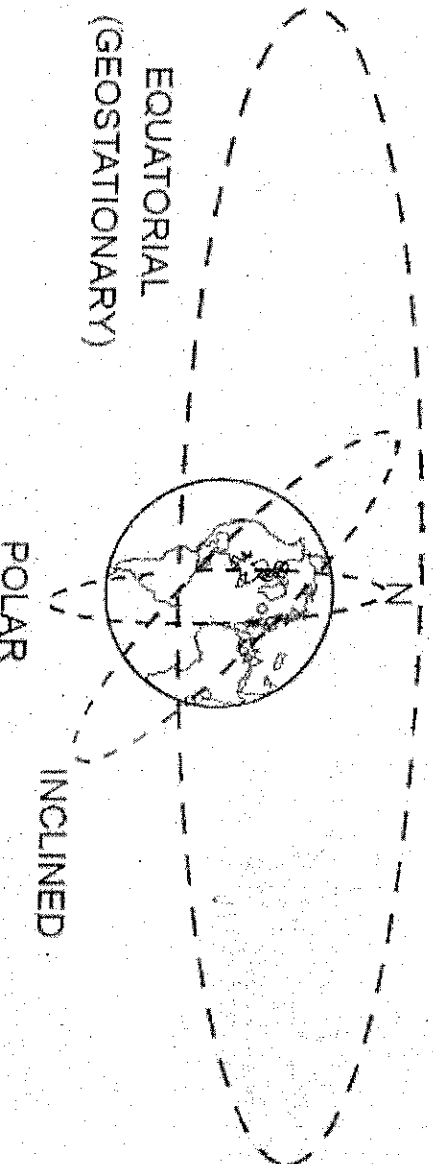
satellites has been expanding steadily over the years since its inception, the available bandwidth is still very small compared to optical fiber capabilities. For speech communications, there is ample available bandwidth, but for video transmission or high-speed data throughput there are severe limitations.

Progress in digital compression techniques is gradually reducing the bandwidth needed for video transmission. Partial-motion videoconferencing is becoming feasible in the kilobit-per-second rather than megabit-per-second realm. Full-motion video, however, still requires several megabits per second, which is fine for broadcasting a few simultaneous TV programs but a problem for the individual requiring multimedia interaction facilities.

On the commercial side, one of the interesting aspects of satellite communications is that the cost of a single-satellite-hop telephone call is *almost independent of distance*. Whether calling a next-door neighbor or someone on another continent, the amount of equipment involved in the process is almost the same. The application determines the type of satellite system required.

- Satellites can be placed in three types of orbit:
1. Equatorial (geostationary)
 2. Polar
 3. Inclined

The type of satellite system determines the altitude at which the satellite is fixed. Figure 1 distinguishes between the three types. The geostationary orbit is the style most widely used for broadcasting, where the satellite is in an equatorial orbit and appears to be at a fixed point in the sky relative to an observer on the earth. Simple Newtonian mechanics shows the altitude of geostationary satellites to be an amazing 36,000 to 41,000 km, depending on the earth latitude of the observer. This vast distance makes life difficult for the



designer, but good-quality voice and video communications via satellite have been commonplace for many years now. One of the major drawbacks of this large distance is the time taken for a round-trip signal to travel from the earth up to the satellite and back down to earth. Even at the speed of light the signal takes 0.24 to 0.27 s. This time delay can cause an annoying echo unless electronic circuitry is employed to minimize the effect. Another disadvantage of the geostationary orbit is the fact that at latitudes farther away from the equator, the geostationary satellite appears lower and lower toward the horizon. Eventually, at about 5° north or south, the satellite is too low on the horizon to receive a clear signal.

Satellites in polar or inclined orbits do not have this problem, but they are no longer geostationary. This would be a problem for broadcasting because an observer on earth would have to track the satellite as the earth moved beneath it, and if permanent transmission were

required, several satellites would be needed with some mechanism for making a smooth transition between them before each one sequentially disappeared over the horizon.

Polar orbits of about 800-km altitude are convenient for global coverage, so long as several satellites are moving in the same polar orbit and several polar orbits are used. These low-earth-orbit (LEO) satellite systems are the subject of potentially revolutionary constellations of satellites that are effectively global cellular systems. Inclined orbits of about 10,000 km provide global coverage with fewer satellites than LEO systems. These medium-earth-orbit (MEO) systems are consequently in direct competition with the LEO systems.

The satellite was traditionally used in the "bent pipe" mode. As the term suggests, the satellite acts like a slingshot to redirect the incoming signal to different locations on earth. In reality, as the simplified satellite link in figure 2.

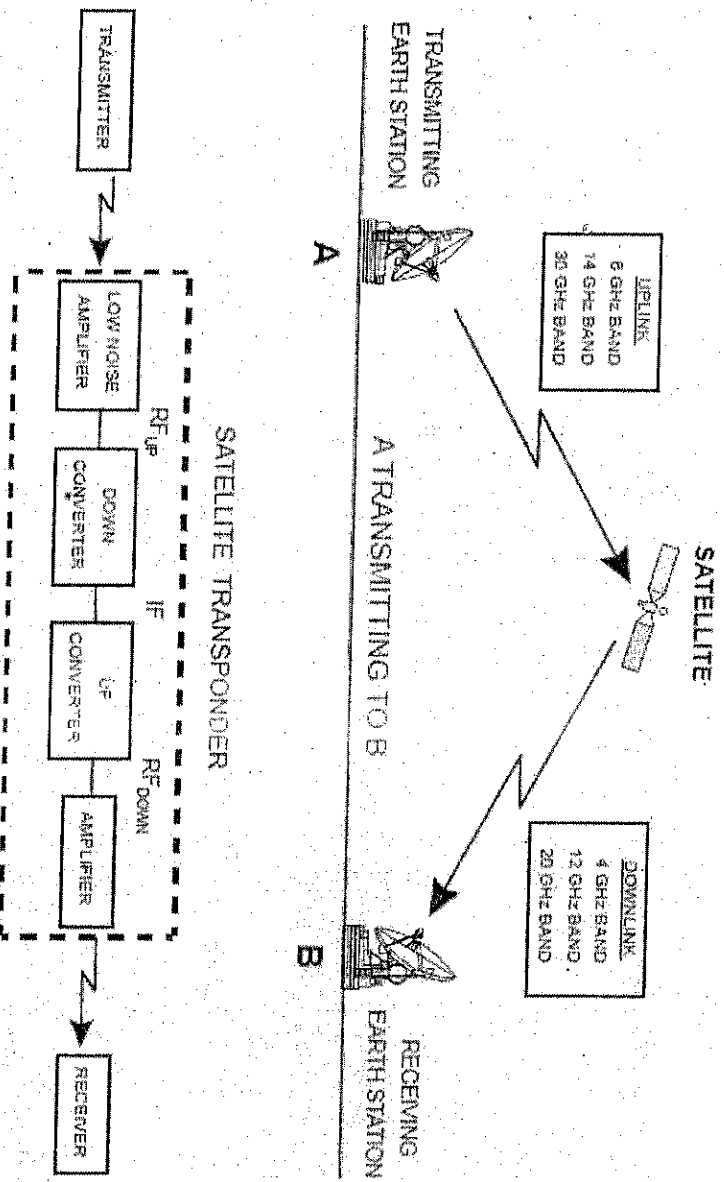


Figure 2 The Satellite Link

indicates, the uplink microwave RF signal is amplified, down converted to the IF, up converted to a different frequency downlink microwave RF signal, amplified again, and retransmitted. That process is hardly the action of a bent pipe. However, this technique has been used since satellite communications began, and is still used on many of today's satellites. The addition of onboard processing in modern satellites allows separate parts of the incoming information package to be routed to different destinations by retransmission via different antenna beams.

1.2 Frequency Allocation

The world has been split up into three regions as indicated on the ITU map. This is accompanied by the uplink and downlink frequency allocations for fixed satellite communications as in Table 1. The main bands of interest are usually referred to by their popular names, which were allocated more than 50 years ago. They are summarized as follows:

Table 1 Frequency Allocation Table

Frequency range (GHz)		
Uplink	Downlink	Frequency band
6	4	C-BAND
14	12	KU-BAND
30	20	KA-BAND

The traditional satellite bands have used the radio window of 1 to 10 GHz, which is relatively free from ionospheric absorption and rain attenuation. at frequencies above 10 GHz. Satellite communications at the Ka-band (30/20 GHz) frequencies have recently started to gain more attention as the C- and Ku-bands have become congested. At the uplink frequency of 30 GHz, an attenuation of about 15 dB is experienced for rain falling at 25 mm/h. In tropical regions this would be a serious problem.

1.3 Geostationary Satellite Communications

GSO satellites orbit in the same rotational direction as the earth at an altitude of about 36,000 km above the equator as calculated by simple Newtonian mechanics. Interestingly, that altitude is the same regardless of the mass of the satellite. Because of its high altitude, a GSO satellite is visible from about 42 percent of the earth's surface. A satellite that orbits at an altitude less than 36,000 km above the equator has to move at a higher velocity to remain in orbit which means the satellite moves relative to the earth. For the orbiting altitude of 36,000 km, the velocity is 3.1 km/s, and for a 780-km altitude (LEO) the velocity is 7.5 km/s. Clearly, any nonequatorial orbit regardless of altitude will cause the satellite to move relative to the earth; that is, the satellite will be in a nongeostationary orbit.

1.4 Satellite parameters

The geostationary satellite is composed of several subsystems, and a view of INTELSAT VIII is shown in Fig. 3.

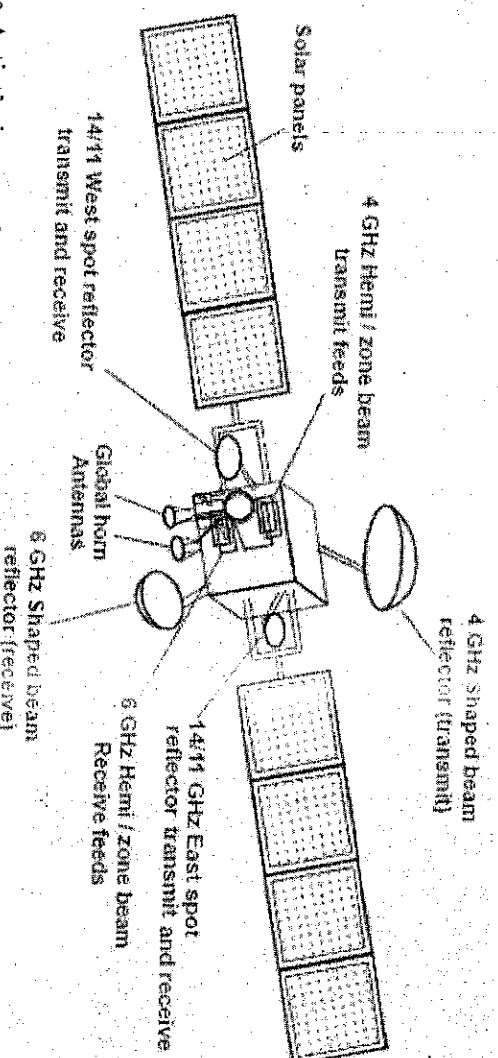


Figure 3 Artist's impression of INTELSAT VIII satellite structure
Major parts of the INTELSAT VIII satellite structure

- An antenna subsystem for receiving and transmitting signals.*
- Transponders, which receive signals, amplify them, translate their frequency, and retransmit them back to earth. In some cases demodulation to baseband, switching, rerouting, and remodulation functions are also included.*
- A power generating and conditioning subsystem to provide power for the satellite and to supply all of the different voltages and currents the electronic circuits require.*
- A command and telemetry subsystem for transmitting data about the satellite to earth and receiving commands from earth.*
- A thrust subsystem for adjusting the satellite orbital position and attitude (orientation in space relative to the earth).*
- A stabilization subsystem to keep the satellite antennas pointing toward earth.*

Assignment

Sketch the frequency spectrum for a satellite indicating its uses

2.0 INFORMATION SUPERHIGHWAY

Defn. A hypothetical electronic communications network encompassing computer networks, television, telephones, etc. The concept of a worldwide network of computers capable of transferring all types of digital information at high speed

The information superhighway brings together millions of individuals who could exchange information with one another

Other definition a telecommunications infrastructure or system (as of television, telephony, or computer networks) used for widespread and usually rapid access to information called also infobahn,

2.1 Telecommunications Networks and their Scope

The electronic transmission of information over distances, called telecommunications, has become virtually inseparable from computers: Computers and telecommunications create value together.

2.2 Components of a Telecommunications Network

Telecommunications are the means of electronic transmission of information over distances. The information may be in the form of voice telephone calls, data, text, images, or video. Today, telecommunications are used to organize more or less remote computer systems into telecommunications networks. These networks themselves are run by computers.

A telecommunications network includes the following components:

1. *Terminals* for accessing the network
2. *Computers* that process information and are interconnected by the network
3. *Telecommunications links* that form a channel through which information is transmitted from a sending device to a receiving device.
4. *Telecommunications equipment* that facilitates the transmission of information.
5. *Telecommunications software* that controls message transmission over the network.

2.3 Scope of Telecommunications Networks

Two principal types of telecommunications networks can be distinguished from the point of view of their geographical scope. They are:

1. Local area networks
2. Wide area networks

Local area network (LAN): is a privately owned network that interconnects processors, usually microcomputers, within a building or on a campus site that includes several buildings.

Characteristics of a LAN

- a. LANs are the principal tool of workgroup computing
- b. LANs ensure high-speed communication within a limited area and enables the users to share facilities (peripherals) connected to it.
- c. Usually include a large-capacity, secondary storage device, where database and applications software are maintained, managed by a microcomputer acting as a file server that delivers data or program files to other computers.
- d. Facilities (peripherals) may include jukebox optical memory and fast printers
- e. Frequently, one of the facilities (peripherals) in a LAN is the gateway hardware and software that give the network users access to other networks.

- f. More group members may connect to the network from remote sites using wireless telecommunications.
- g. Links and equipment of LANs are owned by the user company, and these networks are generally much faster than WANs.
- h. LANs are generally composed of a network of microcomputers

2.4 Wide area network (WAN): is a telecommunications network that covers a large geographical area.

Characteristics of a WAN

- a. The information system of an entire organization may be structured as a hierarchy. The WANs system architecture looks very much like an organization chart.
- b. WANs connect all the divisional minicomputers to the headquarters mainframe with a variety of local microcomputers and terminals located at remote sites connected, in turn, to the minicomputers.
- c. WANs provide the backbone through which all other nodes (computers and terminals) communicate.
- d. WANs often use telecommunication links and equipment provided by specialized vendors, called *common carriers*.
- e. WANs serve to interconnect multiple LANs and can make specific resources available to a large number of workstations.

Metropolitan Area Networks (MAN) - are telecommunications networks that interconnect various local area networks within a metropolitan area, that is, within approximately a 50-mile range.

Characteristics of a WAN

- a. Purpose of MANs is to interconnect various LANs within a metropolitan area, that is, within approximately a 50 – mile range.
- b. Generally, the speed of MANs is equal to that of LANs and they use similar technology.

3.0 Telecommunications Links

Telecommunications links may be implemented with various communication media, with a corresponding variety of characteristics. The main feature of a medium is its potential transmission speed, also known as channel capacity, which for data transmission purposes is expressed in bits per second (bps). An alternative measure of transmission channel capacity is bandwidth - the range of signal frequencies that can be transmitted over the channel.

Six potential media are employed to implement telecommunication links:

- 1. Twisted pair
- 2. Coaxial cable
- 3. Fiber Optics cable
- 4. Terrestrial Microwave
- 5. Satellite Transmission
- 6. Radio Transmission

Three of the above transmission media are classified as guided media - in which the signal moves along an enclosed path. Guided media require wiring. They include:

- 1. Twisted pair
- 2. Coaxial cable
- 3. Fiber optic cable

Three of the below transmission media are classified as wireless media - the signal is broadcast (radiated in many directions) over the air or space and received through an antenna. They include:

1. Terrestrial Microwave
2. Satellite Transmission
3. Radio Transmission

3.1 Characteristics of Communications Media

- a) **Twisted Pair** a communications medium consisting of a pair of wires.
- b) **Coaxial Cable** a communications medium that consists of a relatively thick central conductor shielded by several layers of insulation and the second conductor just under the cable's shell
- c) **Fiber Optics** high-capacity communications medium that consists of many strands of pure glass with a data carrying core in the middle, surrounded by a reflective coating and a protective sheath.
- d) **Terrestrial Microwave** long-distance telecommunications by means of microwave signals travelling on the surface of the earth.
- e) **Satellite Transmission** form of microwave transmission in which the signal is transmitted by an earth station to a satellite which rebroadcasts the signal to the receiving station.
- f) **Radio Transmission** wireless communications technology that transmits voice or data over the air using a lower frequency band than microwaves.

Note: Transmission speeds keep on rising, particularly in the fiber optics area. We are now moving toward a global infrastructure of gigabit-speed fiber optic links relying on digital transmission. In this multimedia environment, data, text, voice, images, and video will travel at speeds of billions of bits per second.